

Predictive Analytics: Soccer Performance Data and All-SEC Honors

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Outline

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Introduction

Introduction: All-SEC

What does All-SEC mean?

- ▶ Recognizes the most outstanding women's soccer athletes during the SEC regular season including:
 - ▶ First-Team All-SEC
 - ▶ Second-Team All-SEC
 - ▶ Third-Team All-SEC
- ▶ Voted on by the league's 16 coaches

Introduction: Wyscout

Wyscout

- ▶ The world's biggest library of soccer video and data
 - ▶ Purchased by Hudl in 2019
- ▶ Provides performance data to every SEC team
 - ▶ Team-level data
 - ▶ Player-level data

Goals	↓ Total	Avg. per 90	xG
	8	0.78	6.61
	7	0.79	5.55
	6	0.64	1.23
	6	0.60	3.68
	5	0.73	1.30
	5	0.59	2.21
	5	0.44	4.45
	4	1.00	2.39
	4	0.89	3.08
	4	0.60	2.31

Introduction: Research Question

Does placing in the top 10
of SEC performance
categories



All-SEC honors?

Data

Data

Table 1: Data Sources

Team Rosters	Wyscout	All-SEC
Name	Goals per 90 mins	First-Team All-SEC
Year	...	Second-Team All-SEC
Team	...	Third-Team All-SEC
Position	Shots Blocked per 90 mins	All-Freshman Team

Note. Team rosters were scraped using the *rvest* package in R.

Data

Data Overview	
DETAILS	
allSEC	<factor> 0, 0, 0, 0, 0
id	<character> cb10a33c-900d-4cb7-b38d-3a76de9f0e8b,...
year	<factor> 2023, 2023, 2023, 2023, 2023
team	<factor> Alabama, Alabama, Alabama, Alabama, A...
pos	<factor> M, D, D, D, D
goalsPer90	<numeric> 0, 0, 0, 0, 0
assistsPer90	<numeric> 0, 0, 0, 0, 0
shotsPer90	<numeric> 0, 0, 0, 0, 0
keyPassesPer90	<numeric> 0, 0, 0, 0, 0
touchesBoxPer90	<numeric> 0, 0, 0, 0, 0
defDuelsPer90	<numeric> 0, 0, 0, 0, 0
aerialDuelsPer90	<numeric> 1, 0, 0, 0, 0
recoveriesPer90	<numeric> 1, 0, 0, 0, 0
interceptionsPer90	<numeric> 1, 0, 0, 0, 0
slidingTacklesPer90	<numeric> 0, 0, 0, 1, 0
shotsBlockedPer90	<numeric> 0, 0, 0, 0, 0

Positional Breakdown		
	N	ALLSEC
Total	1283	74
D	385	16
M	377	24
F	373	23
Multiple	148	11

Methods

Methods

$$\log \left(\frac{P(\text{allSEC} = 1)}{1 - P(\text{allSEC} = 1)} \right) = \beta_0 + \sum_{j=1}^{13} \beta_j X_j$$

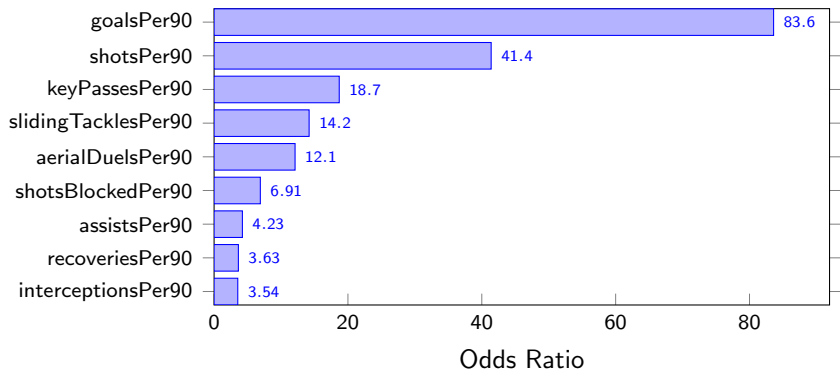
$X_j = \{\text{year, pos, goalsPer90, assistsPer90, shotsPer90,}$
 $\text{keyPassesPer90, touchesBoxPer90, defDuelsPer90,}$
 $\text{aerialDuelsPer90, recoveriesPer90, interceptionsPer90,}$
 $\text{slidingTacklesPer90, shotsBlockedPer90}\}$

Model:

- ▶ Elastic net logistic regression model
- ▶ Penalty tuned to mixture = 0.75
- ▶ Five-fold cross validation
 - ▶ Tuned using Cohen's Kappa
 - ▶ Stratified on allSEC

Results

Results



Results

Model performance:

- ▶ $\text{Kappa} = 0.48$
- ▶ $\text{Sensitivity} = 0.44$

Confusion Matrix		
	NULL	ALL-SEC
Null	244	3
Predicted All-SEC	4	6

Figure 1: Test data

Discussion

Discussion

Attacks reign supreme:

- ▶ Top three most influential features were attacking stats
- ▶ Being amongst the top 10 of goals (per 90 mins) increases your chance of being selected to an All-SEC team by 84%!

Model is not a great predictor of All-SEC awards:

- ▶ Missed three All-SEC players
- ▶ Misidentified four null players

Discussion

Limitations:

- ▶ Wyscout data only provided for players who finished in the top 10 of the respective category
- ▶ Wyscout data was in a .pdf form which could negatively impact the validity of the data
- ▶ Variables were chosen based on prior knowledge
- ▶ No unique identifiers for similar players
- ▶ Only generalizable to women's soccer All-SEC teams

Future research:

- ▶ Include performance data for ALL athletes
- ▶ Cross-validate variables to increase validity and predictive power
- ▶ Increase sample size (number of years)

Thank you!